Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A Manufacturing Foundation AI Model: Learning Dynamic Structural Mechanics and Generation of Optimal Pre-process Setup Solutions			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robotics/Automation Machinery Software	Industrial/General Machinery	Industrial/General Machinery Software
Concept of Proposed Technology	Our research proposes a **"Manufacturing Foundation AI Model"** designed to address the complex processes and conditions inherent in high-variety and low-volume production lines. In job-shop manufacturing systems, achieving process stability and meeting rigorous performance criteria—such as precise dimensions, stiffness, and structural integrity—requires customized processing conditions—such as fixture layouts, tool types and kinematic constraints—for each unique product design. However, identifying optimal process strategies is hindered by the extensive considerations for high-variety components and the high computation costs of traditional Finite Element Method (FEM) simulations. This project aims to develop a Manufacturing Foundation AI Model that			

	comprehends the fundamental manufacturing mechanisms of workpieces under dynamic force loading and generates optimal process conditions in FEM simulations, ensuring robust manufacturability and performance criteria compliance across various product designs. The key machine learning approaches are: **1) Graph Neural Network (GNN):** The model utilizes GNNs to encode 3D mesh data of various machinable geometries of workpieces from FEM, allowing it to effectively learn the physical interactions between the body and dynamic forces. This network functions as a surrogate model, providing guidance throughout the mechanism learning and generative process. **2) Physics-Informed Neural Network (PINN):** To ensure the model's reliable and consistent learning across various product designs and dynamics loadings, we integrate PINNs that embed continuum mechanics principles into the learning objective, allowing for physics-consistent and reliable learning. **3) Flow Matching Algorithm (Generative AI):** Rather than using a traditional probabilistic sampling strategy, we employ Flow Matching Algorithm to learn a deterministic velocity field that maps initial random samples to optimal solutions. By integrating this generative AI model with a trained GNNs, this approach ensures a faster inverse design of optimal process setup conditions compared to traditional optimization processes that only use FEM. We anticipate that this foundation model will serve as a cornerstone for job-shop manufacturing systems trying to transition toward intelligent manufacturing platforms and smart factory ecosystems. By enabling domain-specific customization, this model helps companies reduce simulation-driven lead times and operational costs with minimal manual intervention, thereby enhancing productivity in complex production environments.
Rationale for Joint R&D Project	This joint R&D project is expected to successfully address two main problems: **1) Economic sustainability of Korean manufacturing SMEs amid demographic shifts** and **2) Technology adoption barriers in job-shop manufacturing environments.** 1) Korea's economic growth has been driven primarily by its manufacturing technologies. However, Korean SMEs in the manufacturing sector are currently facing critical challenges: declining price competitiveness in the global market, technical knowledge transfer gaps driven by rapid workforce aging and workforce replacement difficulties. These converging demographic and economic pressures demand a fundamental shift from labor-intensive processes to intelligence-driven systems to recover global market competitiveness and sustain long-term viability of Korean manufacturing SMEs. 2) Mass production systems have been successfully adopting in-line automation and smart factory technologies, benefiting from their standardized processes that require minimal flexibility and a low level of intelligent decision-making. In contrast, make-to-order manufacturers—characterized by high-variety, low-volume production—still struggle with implementing similar technologies due to their

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	inherent complex environments and customized production requirements that existing automation approaches have yet to adequately address. This joint R&D project addresses these problems by leveraging the University of Toronto's world-leading AI expertise and robust computational infrastructure. Through collaboration with UofT, this initiative aims to integrate cutting-edge machine learning models into practical and deployable manufacturing environments. This partnership not only delivers solutions for make-to-order manufacturing challenges but also provides Korean SMEs with a strategic gateway to thrive in the global market through AI-driven solutions.
Key Words	1. Manufacturing Foundation AI Model 2. Machine Learning 3. Smart Factory 4. Manufacturing Mechanisms 5. Generation of Optimal Pre-process Conditions

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (2)	End of the Project	Level (7)
R&D Objectives	The primary objective of this project is to develop a **"Manufacturing Foundation AI Model"** that enables intelligent and automated process planning across a broad range of manufacturing contexts. Unlike previous task-specific research limited to individual processes or narrow product ranges, this project focuses on building a generalizable AI model that learns manufacturing mechanisms and can be integrated into make-to-order manufacturing systems. To achieve this objective, we specifically aim to develop: **1) A Scalable AI Training Framework for Manufacturing Mechanisms Learning** This training framework enables the model to generalize across diverse manufacturing scenarios. ● **Multi-Process Operations:** Learning universal dynamic force mechanisms applicable to milling, cutting, drilling, and assembly operations within a unified framework ● **Material-Aware Deformation Mechanics:** Learning how different materials respond to identical loading conditions ● **Geometry-Invariant Adaptability:** Learning structural mechanisms across various 3D product geometries—from simple prismatic parts to complex topological shapes **2) A Reliable Generative Model for Inverse Design of Optimal Solutions** This framework ensures that the model produces robust and high-fidelity inverse designs of optimal process setup configurations. ● **Robustness in Unseen Scenarios:** Reliable generation performance in extrapolation domains—including new product designs and loading conditions—by adhering to fundamental physical principles rather than relying solely on training data patterns ● **High-Fidelity Generation Outcomes:** Consistent optimal solution generation with less uncertainty through deterministic approaches. The scalability of the training framework addresses the fundamental challenge driven by complex processing environments that make-to-order manufacturers are facing, enabling them to manage broader product portfolios and respond rapidly to custom orders. Furthermore, by producing robust and deterministic inverse design of pre-process setups, manufacturers significantly reduce manufacturing defects and setup errors, enhancing productivity through lower operational costs and minimized production delays which are critical factors for maintaining profitability and competitiveness in the global market.			

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R&D Target Specifications	To validate the Manufacturing Foundation AI Model's performance, we establish two categories of quantitative target specifications: **1) FEM simulation environments** and **2) real-world manufacturing comparison**. **1) Performance Evaluation in FEM Simulation Environment** **Test Dataset:** 100 different geometries across multiple materials and operations. ● **Displacement Prediction Accuracy:** The model accurately predicts workpiece deformation with Normalized Root Mean Square Error below 10% compared to FEM ground truth across all test geometries. $$Norm_RMSE_{displacement}: \varepsilon_u = \frac{\sqrt{\frac{1}{N}\sum_i (u_{i,pred} - u_{i,FEM})^2}}{\sqrt{\frac{1}{N}\sum_i (u_{i,FEM})^2}} \leq 0.1$$ ● **Stress Field Prediction Accuracy:** The model reliably captures stress distribution patterns with high correlation to FEM simulations, ensuring accurate identification of critical stress regions. $$Correlation\ Coefficient: R = \frac{(\sigma_{pred} \cdot \sigma_{FEM})}{\|\sigma_{pred}\| \cdot \|\sigma_{FEM}\|} \geq 0.8$$ ● **Configuration Generation:** The generative model produces pre-process setup configurations that satisfy target deformation criteria in displacement and stress field evaluation. $$Success\ Rate: \mathrm{P}_{sim}(\varepsilon_u \leq 0.1) \geq 0.8$$ **2) Real-World Manufacturing Performance Evaluation** **Test Dataset:** 10 different real-world manufacturing scenarios at 2-3 pilot SME facilities. ● **Comparative Benchmark for Setup Configuration Generation:** AI-generated configurations achieve equivalent or lower dimensional error ε_{dim} compared to expert manual planning in dimensional accuracy evaluations. $$Success\ Rate: \mathrm{P}_{real}(\varepsilon_{dim,expert} \geq \varepsilon_{dim,AI}) \geq 0.7, where\ \varepsilon_{dim} = \frac{|dim_{pred} - dim_{real}|}{|dim_{real}|}$$
R&D Plan	**1. Synthetic Data Generation (0~18 Month)** To establish a comprehensive synthetic dataset of manufacturing scenarios, ANSYS LS-DYNA [1]—a powerful FEM-based multi-physics solver—is leveraged to design a systematic sampling strategy that captures the diversity inherent in make-to-order manufacturing environments: ● **Material Coverage:** The dataset encompasses several primary material families commonly used in machining processes—such as steel alloys and aluminum alloys, enabling the model to learn how different materials respond to identical loading conditions. ● **Geometric Diversity:** We employ modified primitive geometric shapes—T-shape plates, cylinder, rectangular hollow box structure—to learn geometry-invariant basic structural and material removal mechanics applicable to arbitrary product designs. ● **Process Parameter Variation:** Dynamic loading conditions are systemically varied to reflect real manufacturing operations. Tool paths encompass diverse trajectories—including linear cuts, pocket milling, and contour machining—with tool types varying from end mills to drills. ● **Initial Setup Condition:** Initial fixture configurations span from minimal constraint setups to redundant constraint systems with varying spatial distributions. Fixture types range from

fully constrained to partially constrained for the model to learn the different support and fixing strategies that influence the machining stability and outcomes.

2. Data Structure and Feature Extraction (0~18 Month)

Each FEM simulation generates time-series data that captures the complete manufacturing process evolution. As illustrated in ***Figure 1*** [10], the extracted features at each time step are:

- **Node-Level data includes:** Node ID V_n, Coordinates C=(x,y,z), Displacement $u=(u_x,u_y,u_z)$, Velocity $v=(v_x,v_y,v_z)$, Degree of Freedom $\delta=(\delta_x,\delta_y,\delta_z,\delta_{\theta x},\delta_{\theta y},\delta_{\theta z})$, Boundary Condition BC=(0:free, 1:fixed, 2:nodal_force), Applied Force $F=(F_x,F_y,F_z)$, Active State S=(0:Removed, 1:Present)
- **Element-Level data includes:** Edge ID E_n, Connectivity ($e_{12},e_{13},e_{23}\ldots$), Material Properties (E:young's modulus, ν:poisson's ratio, ρ:density, σ_y:yield strength), Residual Stress (σ_{res})
- **Data Format:** The raw output files from FEM are converted to a machine learning-ready format, JSON.

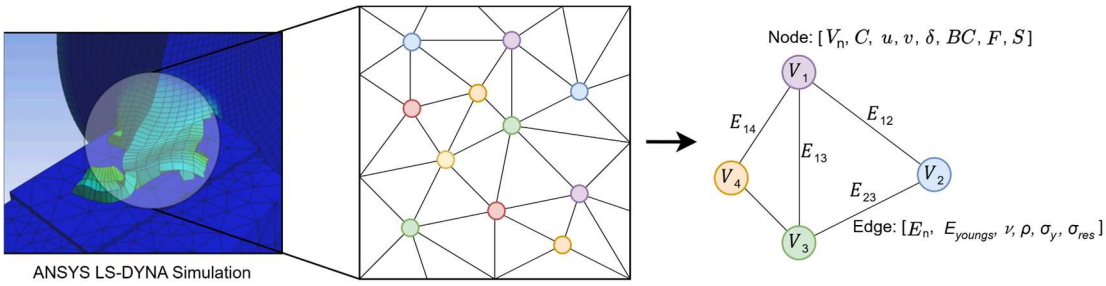

Figure 1. *Graph Representation of FEM Mesh Data*

3. Model Architecture (6~32 Month)

Each FEM simulation generates time-series data containing node and edge features. This dataset is utilized to train a Graph Neural Network (GNN)-based [2, 3] foundation model following the Encoder-Processor-Decoder architecture inspired by MeshGraphNets (MGN) [4, 5], with extensions for physics-informed learning and generative inverse design capabilities.

3.1 Graph Network Construction from Time-Series Data

Graph-based learning demonstrates strong scalability advantages as MGN naturally handles variable mesh sizes, arbitrary geometries, and dynamic forces. The three key steps of GNNs are depicted in ***Figure 2***:

- **Encoder:** The encoder transforms raw FEM data into high-dimensional embeddings for graph-based learning. Node and edge features are processed through a Multi-Layer Perceptron (MLP) to generate embeddings in a unified latent representation space where the model can learn meaningful manufacturing mechanisms.
- **Processor:** Multiple message-passing layers iteratively propagate information across the mesh structure: edge and node update aggregate information from incident components to capture both local stress concentrations and global structural responses. Node deactivation

masking naturally handles dynamic topology changes during material removal processes.

- **Decoder:** The decoder transforms final node embeddings into displacement predictions $\Delta u=[\Delta u_x,\Delta u_y,\Delta u]$ for each time step, enabling autoregressive rollout that predicts the complete temporal evolution of deformation throughout the manufacturing process.

3.2 Learning Target using Physics-Informed Neural Network

The prediction rollouts from the decoder are enforced to adhere to continuum mechanics principles using PINNs [6]: a constitutive law ($\sigma = C:\varepsilon$), an equilibrium constraint ($\nabla\cdot\sigma + f = 0$), and an energy minimization term ($E_{min} = \int_{\Omega} (I+\nabla_X u)d\Omega$). By integrating these physical laws into the loss function, the model learns to produce predictions that are physically consistent and reliable in extrapolation of unseen geometries and loading conditions.

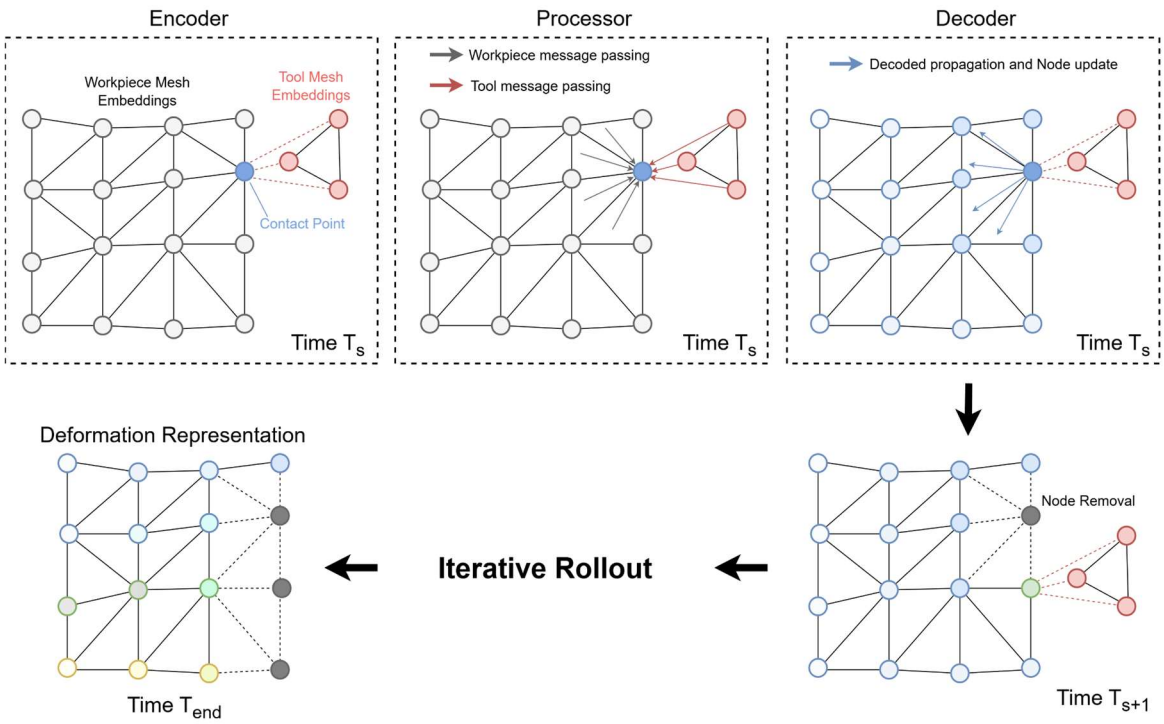

Figure 2. *Diagram of a Graph-based Learning*

3.3 Generative Inverse Design of Optimal Setups using a Flow Matching Algorithm

For generative inverse design of optimal pre-process setup configurations, we employ a Flow Matching algorithm [7, 8, 9], as described in ***Figure 3***, that learns a deterministic mapping (a time-dependent velocity field $v_\theta(z_t,t,c)$) from random initial noise to optimal configuration solutions that minimize the deformation of the remaining workpiece and satisfy quality requirements. This generative approach reduced the time to figure out the optimal pre-process setup in the traditional FEM-based simulation.

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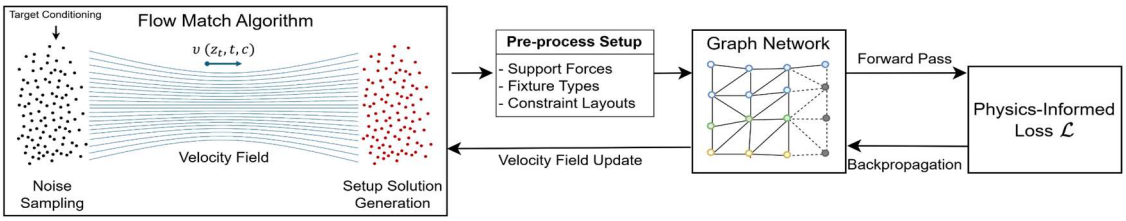

Figure 3. *Inverse Design for Optimal Pre-process Setup Conditions*

4. Customization of The Foundation AI Model (32 ~36 Month)

The foundation AI Model is then tailored through transfer learning for individual manufacturers that operate under unique process conditions and requirements—such as fixture types, precision resolutions, quality standards, etc. This customization step updates only the inverse design generation layers of the model while preserving the underlying mechanisms understandings and enables continuous learning loops based on production feedback to meet the manufacturer's specific requirements.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The University of Toronto will lead the AI model development and provide world-class research expertise and computational infrastructure to enable successful project execution. **1) AI Model Development and Research:** ● Design and development of the Manufacturing Foundation AI Model architecture ● Conduct algorithm research and optimization to enhance the model's performance **2) World-Class AI Expertise and Collaboration:** ● Provide academic supervision and technical guidance from faculty experts at the University of Toronto and Vector Institute—globally leading AI research institutions ● Allocate dedicated Korean graduate student researchers to ensure effective collaboration **3) Computational Infrastructure and Resources:** ● Provide access to SciNet—high-performance computing center—for parallel FEM simulation, large-scale datasets, and distributed model training
Expected R&R of Korean SME Partner	Korean SME partners will provide technical advice, real-world manufacturing data, and facility access essential for validating the model's practical applicability and successful deployment. **1) On-site Manufacturing Data:** ● Provide FEM simulation datasets of actual production parts from their facilities for model testing ● Provide real-world manufacturing data and records of actual pre-process setup conditions from vision systems **2) Technical Support:** ● Share technical specifications, process criteria, and quality standards of their manufacturing operations for model customization ● Provide representative reference components from their product portfolios **3) Support for Real-World Deployment and Test:** ● Cooperate in pilot-scale deployment and real-world performance testing ● Participate in progress reviews and provide feedback for continuous model refinement

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	● **Technological Achievements:** This joint R&D project will deliver a Manufacturing Foundation AI Model capable of intelligent, automated process planning in make-to-order scenarios by demonstrating two key technological breakthroughs: scalability across diverse manufacturing operations and physically reliable performance through physics-informed learning that ensures consistent predictions even for unseen product designs. ● **Commercialization Potential:** Following successful demonstrations of customized deployments in 2-3 real manufacturing facilities, this foundation model will enable commercialization across multiple manufacturers through its transfer learning framework. By introducing a feedback learning pipeline, the tailored model can continuously adapt to their manufacturing processes, resulting in stable and reliable planning performance in the long run. ● **Industrial and Economic Impact:** This technology addresses critical challenges that Korean manufacturing SMEs are facing—such as workforce aging and technical knowledge transfer gaps, and declining price competitiveness. This AI-driven intelligent system will contribute to improvements of productivity through operational efficiency, lead time reduction, and defect minimization in Korean manufacturing industries. Furthermore, this project provides Korean SMEs with an opportunity to strengthen their price competitiveness and capture larger shares of manufacturing sectors in global markets.

6. References

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Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization